

Experiment No: 3

Title:

To perform various GIT operations on local and remote repositories using GIT Cheat-Sheet.

Aim:

To perform basic Git operations such as creating files, staging changes, committing updates, viewing logs, and synchronizing local repositories with remote repositories using Git commands.

Theory:

Git is a distributed version control system that helps developers track changes in source code during software development. It enables multiple users to work on the same project simultaneously without conflicts. Git maintains a history of changes and allows users to revert to earlier versions when required.

Local repositories store project files on a user's system, while remote repositories are hosted on platforms such as GitHub for collaboration and backup. Operations such as add, commit, push, pull, and log help manage and synchronize project files between local and remote repositories efficiently.

Requirements:

- Git Bash installed
 - GitHub account
 - Initialized Git repository
 - Internet connection
-

Procedure:**Step 1:** Create files inside repository

Command:

```
touch index.html  
touch teststatus
```

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ touch index.html
touch teststatus
```

Step 2: Display created files

Command:

ls

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ ls
index.html  teststatus
```

Step 3: Add files to staging area

Command:

git add .

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git add .
```

Step 4: Check repository status

Command:

git status

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.html
        new file:   teststatus
```

Step 5: Commit files to local repository

Command:

`git commit -m "First Commit"`

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git commit -m "First Commit"
[master (root-commit) 5a4bd8f] First Commit
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 index.html
create mode 100644 teststatus
```

Step 6: Modify file and perform express commit

Command:

`nano index.html``git add .``git commit -am "Express commit"`

```
chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ nano index.html

chetna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git add .
git commit -am "Express commit"
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
[master bf2831b] Express commit
1 file changed, 3 insertions(+)
```

Step 7: View commit history

Command:

`git log`

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git log
commit bf2831bf8a79ac59eb62de626f6c82d1a37f4517 (HEAD -> master)
Author: geetmulki <geetmulki@gmail.com>
Date: Mon Mar 9 20:19:57 2026 +0530

    Express commit

commit 5a4bd8f3968b5c0e029978114bcfd59bcc4f3143
Author: geetmulki <geetmulki@gmail.com>
Date: Mon Mar 9 20:19:20 2026 +0530

    First Commit
```

Step 8: View commit history in one-line format

Command:

git log --oneline

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git log --oneline
bf2831b (HEAD -> master) Express commit
5a4bd8f First Commit
```

Step 9: Add remote repository connection

Command:

git remote add origin https://github.com/geetmulki/sepm_73

```
chamna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git remote add origin https://github.com/geetmulki/sepm_73
```

Step 10: Verify remote repository connection

Command:

git remote show origin

```
chenna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git remote show origin
* remote origin
Fetch URL: https://github.com/geetmulki/sepm_73
Push URL: https://github.com/geetmulki/sepm_73
HEAD branch: (unknown)
```

Step 11: Push local repository to remote repository

Command:

git push -u origin master

```
chenna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 504 bytes | 252.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/geetmulki/sepm_73
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Step 12: Pull updates from remote repository

Command:

git pull

```
chenna mulki@LAPTOP-284977H7 MINGW64 ~/Desktop/Geet Mulki/git-dvcs/git-demo-project (master)
$ git pull
Already up to date.
```

Result:

Successfully performed various Git operations such as add, commit, log, push, and pull on local and remote repositories.

Conclusion:

Thus, different Git commands were executed successfully to manage files in both local and remote repositories using version control operations.